An Explicit Quaternionic S-Arithmetic Height Clock: A Scoped Route-A Assessment

Hilbert–Pólya Dynamical Structure Exploration Project

August 7, 2026

Abstract

We test a compact quaternionic S-arithmetic system as a source of a canonical arithmetic clock for a Hilbert–Pólya dynamical determinant. For the quaternion algebra $B = (-1, 3)_{\mathbb{Q}}$, localized at $p = 13$, the projective S-units $\varepsilon = 2 + \sqrt{3}$ and $\pi = 4 + \sqrt{3}$ generate an explicit periodic flat with joint Cartan vector

$$(m, n) \longmapsto (mA + nC, n)\,, \qquad A = 2\log(2 + \sqrt{3}), \quad C = \log \frac{4 + \sqrt{3}}{4 - \sqrt{3}}.$$

The clock has rank two and an exact primitive/repetition law. Its archimedean projection is nevertheless nonproper: continued fractions give primitive directions with $|mA + nC| \to 0$ and $|n| \to \infty$. Hence the real-only primitive class Euler product has no ordinary convergence half-plane, already on this single flat. The same arithmetic supplies a canonical repair. For the eigenvalue ratio r_γ,

$$2h(r_\gamma) = \ell_\infty(\gamma) + (\log 13)\ell_{13}(\gamma),$$

so the absolute Weil height is a proper homogeneous scalar clock. We prove that primitive height directions on the flat have an explicit quadratic counting law. This does not produce a rank-one Ruelle determinant: regular classes lie in compact flat tori and require centralizer/regulator weights. Moreover, the direct self-adjoint compact Laplace–Hecke baseline retains the surface Weyl law $N(T) \asymp T^2$, not the Riemann–von Mangoldt order $T \log T$. The result is therefore an explicit arithmetic-clock example with scoped negative Hilbert–Pólya obstructions, not a general no-go theorem for higher-rank Lefschetz, scattering, conductor-growing, or unbounded height-operator constructions.

1 Introduction

The Hilbert–Pólya search asks for more than a chaotic spectrum. A useful candidate must specify an intrinsic classical clock, a primitive/repetition ledger, an analytic determinant, and a natural self-adjoint spectral object with the Riemann–von Mangoldt counting order. The area-preserving Hénon model that motivated this research track supplies an honest symplectic map and suggestive numerical spectra, but its continuum regularization and effective Planck scale are modeling choices and its GUE statistics are not a determinant identity [12]. Several subsequent finite-state and graph-cover constructions also encountered divisor-density or chronology obstructions. The present work therefore changes the dynamical system rather than modifying another finite transfer matrix.

Our candidate is the compact S-arithmetic quotient associated with a division quaternion algebra that splits at the real place and at one finite prime:

$$\Gamma\backslash(\mathbb{H} \times T_p).$$

It offers three features absent from the elementary Hénon baseline: a genuine non-Archimedean displacement, an arithmetic Hecke correspondence, and a canonical compact automorphic

Hilbert space. Real/p-adic diagonal actions of this type are established objects in homogeneous dynamics [6, 8]. Our question is not whether this theory exists, but whether its most explicit primitive length data survive the Hilbert–Pólya gates.

1.1 Explicit calculation and scoped obstructions

We freeze the algebra

$$B = (-1, 3)_{\mathbb{Q}}, \qquad p = 13,$$

and the quadratic torus $K = \mathbb{Q}(\sqrt{3})$. The two projective S-units

$$\varepsilon = 2 + \sqrt{3}, \qquad \pi = 4 + \sqrt{3}$$

generate the exact centralizer lattice. For $\gamma_{m,n} = [\varepsilon^m \pi^n]$, we prove

$$\ell_\infty(\gamma_{m,n}) = |mA + nC|, \qquad \ell_{13}(\gamma_{m,n}) = |n|, \tag{1}$$

with A, C as in the abstract. The clock determinant is $A \neq 0$, so the product formula does not collapse the two places.

This positive result exposes two obstructions. First, the quotient of the minimal real/tree flat by the rank-two centralizer is a compact flat torus. The primitive group class is canonical, but its geodesics form a parallel family rather than an isolated Ruelle orbit. This is the familiar reason that higher-rank trace formulae use flat and centralizer-volume weights; periodic-flat counting and equidistribution are themselves established subjects [3, 2, 7].

Second, C/A is irrational for an arithmetic reason. Primitive lattice directions approach the real Weyl wall:

$$|m_k A + n_k C| \longrightarrow 0, \qquad |n_k| \longrightarrow \infty.$$

The real-only primitive Euler factors therefore fail even the necessary condition of tending to one. No amount of longer orbit enumeration can fix this failure.

The repair is also exact. If r_γ is the eigenvalue ratio, then

$$H(\gamma) := \ell_\infty(\gamma) + (\log 13)\ell_{13}(\gamma) = 2h(r_\gamma). \tag{2}$$

The coefficient $\log 13$ is not fitted: it is forced by the local normalization in the absolute Weil height. The primitive directions of this height clock satisfy

$$N_H(R) \sim \frac{6}{\pi^2 A \log 13} R^2.$$

Finally, we audit the natural self-adjoint baseline. A normalized Hecke operator at 13 is bounded and commutes with the Laplacian on the compact arithmetic surface. By the min–max principle, any bounded function of that Hecke operator added to the positive Laplace spectral parameter preserves the T^2 Weyl order. It cannot have the $T \log T$ divisor count of the completed Riemann function under a fixed affine spectral map.

1.2 Claim boundary

We do not define a new global higher-rank zeta. Higher-rank Lefschetz/prime-geodesic frameworks, p-adic torus formulae, modular Hecke correspondences and joint Laplace–Hecke asymptotics are prior art [3, 2, 4, 5, 1, 9]. A Hecke–Ruelle construction was also publicly announced in June 2026 [10], but no associated preprint was located at our source lock; we treat it as a priority risk, not a proved result.

Our negative spectral statement is equally scoped. It does not exclude a regulator-weighted higher-rank Lefschetz determinant, a conductor-growing family, a canonical sparse projection, scattering resonances, or a new unbounded operator whose principal symbol is derived from (2).

2 The arithmetic host

Let

$$B = \mathbb{Q}\langle i, j : i^2 = -1,\ j^2 = 3,\ ij = -ji\rangle$$

and $\mathcal{O} = \mathbb{Z}[1, i, j, ij]$. Write $G = \mathrm{PGL}_1(B)$. The reduced norm is

$$\mathrm{nrd}(a + bi + cj + dij) = a^2 + b^2 - 3c^2 - 3d^2.$$

Standard facts about quaternion algebras and their orders used below may be found in [11].

Proposition 2.1. *The algebra B is ramified exactly at the finite places 2 and 3. It is split at infinity and at 13. Consequently*

$$\Gamma = P(\mathcal{O}[1/13]^\times) < G(\mathbb{R}) \times G(\mathbb{Q}_{13}) \simeq \mathrm{PGL}_2(\mathbb{R}) \times \mathrm{PGL}_2(\mathbb{Q}_{13})$$

is a uniform arithmetic lattice.

Proof. For the Hilbert symbol $(-1, 3)_v$, the local values at 2 and 3 are -1. At an odd prime away from 6, both entries are units and the symbol is $+1$. At the real place the second entry is positive, so the symbol is $+1$. In particular it is $+1$ at 13; explicitly, $3 \equiv 4^2 \pmod{13}$. Thus the ramification set is $\{2, 3\}$ and B is a division algebra over $\mathbb{Q}$ but split at the two selected local places. The S-arithmetic lattice theorem applied to the localized order gives the lattice assertion. Uniformity follows from the $\mathbb{Q}$-anisotropy of G. □

Put $K = \mathbb{Q}(j) = \mathbb{Q}(\sqrt{3})$. Its integer ring is $\mathcal{O}_K = \mathbb{Z}[\sqrt{3}]$. Define

$$\varepsilon = 2 + j, \qquad \pi = 4 + j.$$

Then

$$N(\varepsilon) = 1, \quad N(\pi) = 13, \qquad \varepsilon^{-1} = 2 - j, \quad \pi^{-1} = \frac{4 - j}{13}.$$

Thus both projective classes lie in Γ.

Proposition 2.2. *The projective centralizer of $[\pi]$ in Γ is*

$$C_\Gamma([\pi]) = \mathcal{O}_K[1/13]^\times/\mathbb{Z}[1/13]^\times \cong \mathbb{Z}^2,$$

with basis $[\varepsilon], [\pi]$.

Proof. If $[x]$ centralizes $[\pi]$, then $x\pi x^{-1} = q\pi$ for some $q \in \mathbb{Q}^\times$. Reduced-trace invariance and $\mathrm{trd}(\pi) = 8$ give $8 = 8q$, hence $q = 1$. The ordinary centralizer of the regular noncentral element π in B is K, and coefficient comparison gives $\mathcal{O}[1/13] \cap K = \mathcal{O}_K[1/13]$. The field $\mathbb{Q}(\sqrt{3})$ has class number one and unit group generated by -1 and ε. Since

$$13 = (4 + \sqrt{3})(4 - \sqrt{3}) = \pi\pi',$$

every 13-unit is $\pm\varepsilon^m\pi^a(\pi')^b$. Modulo rational 13-units, $\pi' \equiv \pi^{-1}$ and -1 disappears. The two displayed classes are therefore a free basis. □

Write $\gamma_{m,n} = [\varepsilon^m\pi^n]$. The only rational class in this torus is $\gamma_{0,0}$: equality of the valuations at the two primes over 13 first forces $n = 0$, after which $\varepsilon^m \in \mathbb{Q}$ forces $m = 0$. Hence every nonzero pair is regular. Its two real conjugates are positive, so its reduced trace is nonzero; the same trace argument used above shows that its projective centralizer is the displayed K-torus. Projective torsion in the centralizer has been removed, and

$$\gamma_{m,n}^k = \gamma_{km,kn}.$$

Any projective root centralizes the element and therefore belongs to this same projective centralizer. Hence $\gamma_{m,n}$ is primitive in Γ, for $(m,n) \neq (0,0)$, precisely when $\gcd(m,n) = 1$. Finally,

$$N_{B^\times}(K)/K^\times \simeq \mathrm{Gal}(K/\mathbb{Q}),$$

and conjugation by i realizes the nontrivial automorphism. It sends both generators to their projective inverses, so simultaneous sign is the only additional conjugacy identification among these directions.

3 An exact two-place clock

For $a + b\sqrt{3} \in K$, multiplication in the basis $(1, \sqrt{3})$ is

$$M(a,b) = \begin{pmatrix} a & 3b \\ b & a \end{pmatrix}, \qquad \det M(a,b) = a^2 - 3b^2.$$

Set

$$A = 2\log(2+\sqrt{3}), \qquad C = \log\frac{4+\sqrt{3}}{4-\sqrt{3}}.$$

Theorem 3.1 (joint-clock certificate)**.** *After orienting the real axis and the* 13*-adic apartment, the signed Cartan map on the centralizer lattice is*

$$\mathbf{c}(\gamma_{m,n}) = (mA + nC, n).$$

In particular,

$$\ell_\infty(\gamma_{m,n}) = |mA + nC|, \qquad \ell_{13}(\gamma_{m,n}) = |n|,$$

and the clock has rank two, with covolume

$$\det\begin{pmatrix} A & C \\ 0 & 1 \end{pmatrix} = A > 0.$$

Proof. At the two real embeddings, the eigenvalue ratios are

$$\frac{2+\sqrt{3}}{2-\sqrt{3}} = (2+\sqrt{3})^2, \qquad \frac{4+\sqrt{3}}{4-\sqrt{3}},$$

with logarithms A and C. At 13, use the Hensel lifts of the two roots of $X^2 - 3$ congruent to $\pm 4 \pmod{13}$. The components of ε are units. The components of π have valuations 0 and 1. Translation length on the Bruhat–Tits tree is the absolute eigenvalue-valuation difference. Additivity on the torus proves the formula. □

There is a useful invariant check that avoids numerical p-adic diagonalization. At the real place let the element be orientation-preserving hyperbolic with positive reduced norm $N > 0$; at the finite place let it be $\mathbb{Q}_p$-split regular. If its reduced trace is t and $\Delta = t^2 - 4N$, then the respective local formulae are

$$\ell_\infty = 2\,\mathrm{arcosh}\,\frac{|t|}{2\sqrt{N}}, \qquad \ell_p = \max\{0, v_p(N) - v_p(\Delta)\}. \tag{3}$$

Indeed, if the two eigenvalue valuations are $a < b$, then $v_p(\Delta) = 2a$ and $v_p(N) = a + b$; equal valuations give zero projective translation. For ε and π the triples (t, N, Δ) are $(4, 1, 12)$ and $(8, 13, 12)$, respectively.

Remark 3.2. The determinant in theorem 3.1 is the first positive gate. It rules out a fake coupling in which the tree displacement is a fixed multiple of the real length. Global product-formula cancellation occurs between conjugate local components; it does not identify the two place coordinates.

Primitive joint boxes already exhibit the rank-two geometry. Visible lattice-point density gives, as $X, Y \to \infty$ with X/Y bounded away from 0 and ∞,

$$\#\{\gamma_{m,n} : \gcd(m,n) = 1, \ell_\infty \le X, \ell_{13} \le Y\}/\{\pm 1\} \sim \frac{12XY}{\pi^2 A}. \tag{4}$$

The high-precision reproducible counts for $X = Y = 10, 40, 160, 320$ are $48, 742, 11841, 47349$; the last ratio to the main term is $1.001685\ldots$.

4 Periodic flats and near-wall accumulation

4.1 The primitive object is not an isolated orbit

For a bi-hyperbolic class, the minimal set in the product geometry is

$$\mathrm{Min}(\gamma) = \mathrm{Axis}_\infty(\gamma) \times \mathrm{Axis}_{13}(\gamma) \cong \mathbb{R}^2. \tag{5}$$

Proposition 4.1. *The centralizer lattice generated by ε, π acts on* (5) *with translation basis $(A, 0), (C, 1)$. Its quotient is a compact flat torus of area A in the chosen normalization. It maps to the arithmetic quotient as an immersed periodic flat; the full finite Weyl normalizer can further produce an orbifold quotient, such as the torus modulo inversion. In either description, a primitive direction labels a parallel family of closed geodesics rather than an isolated periodic orbit.*

This is not a defect of the example. It is the standard higher-rank geometry behind flat-volume and centralizer-covolume terms in Lefschetz and prime-geodesic formulae, including modern periodic-torus counting and equidistribution [3, 4, 2]. An unweighted product over group classes may have the correct algebraic repetition law, but it is not automatically a Ruelle determinant for the geometric dynamics.

4.2 The real projection is nonproper

Theorem 4.2 (near-wall sequence)**.** *There are coprime integer pairs (m_k, n_k) such that*

$$|n_k| \longrightarrow \infty, \qquad |m_k A + n_k C| \longrightarrow 0.$$

Proof. We first show $C/A \notin \mathbb{Q}$. If $qC = rA$ for nonzero integers, then exponentiation gives

$$\left(\frac{\pi}{\pi'}\right)^q = \left(\frac{\varepsilon}{\varepsilon'}\right)^r.$$

At a prime of K above 13, the left side has valuation $\pm q$, whereas the right side has valuation zero. This is impossible. If p_k/q_k is a continued-fraction convergent to C/A, set $(m_k, n_k) = (-p_k, q_k)$. These pairs are coprime and have the desired error with unbounded denominator. $\square$

Some certified record directions are shown in table 1.

There is no contradiction with compactness: the joint displacement escapes in the tree direction. Only its archimedean projection approaches zero.

Table 1: Primitive directions approaching the real Weyl wall.

(m,n)	ℓ_∞	ℓ_{13}
$(-6,17)$	0.04113913165	17
$(-19,54)$	0.02425898145	54
$(-44,125)$	0.00737883124	125
$(-113,321)$	0.00212248772	321

4.3 Euler-product consequence

Let $\mathcal{P}_F$ be the primitive directions of this flat modulo simultaneous sign. Consider the class product

$$Z_F(s,w) = \prod_{(m,n)\in\mathcal{P}_F} \left(1 - e^{-s|mA+nC|-w|n|}\right)^{-1}. \tag{6}$$

Corollary 4.3. *For every $s \in \mathbb{C}$ with $\Re s > 0$, the specialization $Z_F(s,0)$ has no ordinary infinite-product convergence.*

Proof. Along theorem 4.2, the quantity inside the exponential approaches zero. Hence the inverse local factors in (6) do not approach one. This violates a necessary condition for a nonzero convergent infinite product. □

For real $s, w > 0$, (6) does converge absolutely. Indeed,

$$\sup_{t\in\mathbb{R}} \sum_{m\in\mathbb{Z}} e^{-sA|m-t|} < \infty,$$

and the remaining sum over canonical representatives $n \geq 0$ is bounded by a geometric series in e^{-w}. The same bounds are locally uniform on the positive quadrant, so logarithmic differentiation is valid there and

$$\partial_s\partial_w \log Z_F(s,w) = \sum_{\mathcal{P}_F}\sum_{r\geq 1} r\ell_\infty\ell_{13}e^{-r(s\ell_\infty+w\ell_{13})} > 0.$$

Thus the joint class product is not $F(s)G(w)$. The obstruction is not a trivial factorization into real and p-adic marginals; it is the loss of properness when the second clock is discarded.

5 The canonical Weil-height repair

For $\alpha = \varepsilon^m\pi^n$, define the eigenvalue ratio

$$r_{\gamma_{m,n}} = \frac{\alpha}{\alpha'},$$

which is well defined up to inversion by the projective class. Its two real local logarithms are $x, -x$. With the standard normalized absolute values, its finite divisor is $n(\mathfrak{p}-\mathfrak{p}')$ at the two degree-one primes over 13, so the corresponding local logarithms are $n\log 13, -n\log 13$. All other finite local contributions vanish for the S-units under consideration.

Theorem 5.1 (height identity)**.** *For every $\gamma_{m,n}$,*

$$2h(r_{\gamma_{m,n}}) = \ell_\infty(\gamma_{m,n}) + (\log 13)\ell_{13}(\gamma_{m,n}).$$

Proof. The absolute logarithmic Weil height is the sum of positive local logarithms divided by $[K:\mathbb{Q}]=2$. Exactly one logarithm in each conjugate pair is positive. Therefore

$$h(r_{\gamma_{m,n}}) = \frac{1}{2}\left(|mA+nC| + |n|\log 13\right).$$

$\square$

Define

$$H(m,n) = |mA+nC| + (\log 13)|n|.$$

The roof is homogeneous under repetition. It is also proper: bounded H first bounds $|n|$ and then bounds $|m|$. Thus the product formula does not collapse the joint vector, but the height construction does select a canonical scalarization.

Proposition 5.2 (primitive height count)**.** *Let $N_H(R)$ count primitive pairs modulo simultaneous sign with $H(m,n)\le R$. Then, as $R\to\infty$,*

$$N_H(R) \sim \frac{6}{\pi^2 A\log 13}R^2.$$

Proof. The map $(m,n)\mapsto(x,y)=(Am+Cn,n)$ has determinant A. In the (x,y) plane the height ball is the diamond

$$|x| + (\log 13)|y| \le R,$$

of area $2R^2/\log 13$. Pullback divides this by A. Visible lattice points have density $6/\pi^2$, and quotienting by simultaneous sign divides the result by two. $\square$

Table 2: High-precision primitive height counts and their geometry-of-numbers main term.

R	computed count	main term	ratio
20	36	35.9941	1.00017
80	577	575.905	1.00190
320	9211	9214.48	0.999622
640	36857	36857.92	0.999975

The height identity is a positive arithmetic result, but not a Hilbert–Pólya determinant. It forgets the joint Weyl-chamber direction, and it does not remove the clean periodic-flat multiplicity in proposition 4.1. A correct height zeta must still be placed in a regulator-weighted higher-rank trace framework.

6 The natural self-adjoint baseline has the wrong Weyl order

Let $M=\Gamma_0^+\backslash\mathbb{H}$ be the compact orientation-preserving arithmetic orbifold attached to the unlocalized quaternion order, let $\Delta_M\ge 0$ be its hyperbolic Laplacian, and let T_{13} be the normalized self-adjoint Hecke operator associated with the 13-double coset. The latter is bounded and commutes with Δ_M. This is the most direct spherical spectral shadow of the real/tree quotient. Joint Hecke eigenvalue asymptotics in the Laplace aspect on compact arithmetic quotients form an established theory [9].

Theorem 6.1 (bounded-Hecke Weyl obstruction)**.** *For a bounded real Borel function f and $b\in\mathbb{R}$, set*

$$D_{b,f} = \sqrt{\Delta_M+1} + bf(T_{13}).$$

Then $D_{b,f}$ is self-adjoint with compact resolvent and

$$N_{D_{b,f}}(T) := \#\{\lambda_j(D_{b,f}) \le T\} \sim \frac{\mathrm{area}(M)}{4\pi} T^2.$$

Eigenvalues are counted with multiplicity.

Proof. The Hecke term is bounded and self-adjoint, so bounded-perturbation theory preserves self-adjointness and compact resolvent. Put $C = |b|\,\|f(T_{13})\|$. The min–max principle bounds the eigenvalues of $D_{b,f}$ within C of those of $\sqrt{\Delta_M + 1}$. Equivalently, the counting function is squeezed between the unperturbed counts at $T - C$ and $T + C$. The two-dimensional Weyl law for Δ_M gives the result. □

The completed Riemann function has nontrivial-zero count

$$N_\xi(T) = \frac{T}{2\pi} \log \frac{T}{2\pi} - \frac{T}{2\pi} + O(\log T).$$

No fixed affine rescaling converts the T^2 asymptotic in theorem 6.1 into this $T \log T$ law. Adding a bounded Hecke label to a compact-surface spectrum therefore does not solve the spectral-density gate.

Remark 6.2 (what the theorem does not say)*.* The theorem concerns the full spherical spectrum and bounded functions of a fixed Hecke operator. It does not exclude a canonically defined sparse spectral projection, a family whose conductor grows with T, a scattering construction, or an unbounded operator derived from the height clock. Each would need its own domain, compactness/scattering statement and trace identity. Choosing such an object after inspecting Riemann zeros would not be a natural lift.

There is a further chronology caution. Powers T_{13}^n count radial paths with backtracking; they are not the same as a primitive nonbacktracking translation of tree length n. A valid Hecke-return trace must retain this distinction. Replacing the chronological tree dynamics by a simple averaged transition operator would discard precisely the data the joint clock was introduced to preserve.

7 Route-A assessment and system switch

The candidate passes one unusually strong structural gate: it has a canonical, exact, independent arithmetic clock. It then fails the rank-one Hilbert–Pólya route for reasons that are analytic and geometric rather than numerical.

Layer	Verdict	Decisive reason
A1 primitive orbits	A1 weak	Primitive roots and repetitions are exact, but the geometric fixed set is a periodic flat torus, not an isolated orbit.
A2 dynamical zeta	A2 fail	The real-only class product diverges along primitive near-wall classes; a correct global regulator-weighted determinant is not constructed.
A3 analytic structure	A3 fail	No completed divisor/functional equation is derived, and the natural compact spectrum has T^2 rather than $T \log T$ counting.
A4 liftability	A4 fail	The Laplacian and Hecke operator are adjacent canonical objects, but no lift generates or preserves the Weil-height clock; the bounded baseline also fails the counting gate.

The overall decision is a **scoped Route-A rejection**. This is an internal stop decision, not a protocol-complete positive evaluation: no global orbit enumeration, stability multipliers, phases, Fredholm determinant, divisor computation, or shuffled-orbit controls exist for this flat-family candidate. The exact reusable design rules are:

1. A local-place projection of an S-unit clock need not be proper even on a compact quotient.
2. The product formula can preserve a genuinely multidimensional clock; the absolute Weil height may then give a canonical proper scalarization.
3. In higher rank, primitive group classes and primitive isolated dynamical orbits are different data types. Regulator/clean-fixed-set weights cannot be silently omitted.
4. A bounded Hecke correspondence labels a compact surface spectrum but does not change its leading Weyl dimension.

The result does not justify further large orbit sweeps in this compact baseline. The next system class should therefore change the spectral mechanism: a noncompact arithmetic scattering system with a countable-state chronological return map is the preferred direction. It must first pass a strict novelty test against classical modular scattering, where Riemann zeta ratios are already known.

A Reproducibility certificate

The repository contains a standard-library Python implementation. Exact `Fraction` arithmetic is used for multiplication in $\mathbb{Q}(\sqrt{3})$, reduced trace, norm, discriminant, matrix determinant, and 13-adic valuations. The producer uses 80-digit `Decimal` arithmetic for real logarithms, near-wall selection, finite boundary decisions, and asymptotic predictions; binary floats are used only to serialize compact diagnostics. The code reads neither prime tables nor Riemann zeros.

The producer creates:

- `exact_certificates.json`, containing the arithmetic host, generators, joint clock, exact sample invariants and all compact tables;
- `near_wall.csv`, the continued-fraction record sequence;
- `primitive_box_counts.csv` and `primitive_height_counts.csv`;
- a four-artifact internal-consistency hash manifest.

The checker imports no producer code. It enforces exact schemas and required row sets, rederives every released arithmetic and numerical field, uses a separate 120-digit brute-force enumeration and an independent computation of π, and compares embedded JSON tables with CSV output. Sixteen grouped assertions pass with an empty error list. The eleven-test suite includes a fresh temporary-directory reproduction and mutations for missing, empty, duplicated, and altered artifacts. Agreement at 80 and 120 digits together with positive boundary margins makes the finite tables robust; it is not a directed-interval or transcendence proof.

From the project directory, run

```
python code/s_arithmetic_clock.py --output results
python code/independent_check.py --results results \
  --output results/independent_check.json
(cd code && python -m unittest -v test_s_arithmetic_clock.py)
python code/release_manifest.py --verify
```

The hardened computation source and generated results are frozen at commit `24553c8`. The complete documentation, evaluation, manuscript, PDF, and independent-check output are covered by the repository release manifest and Git tag `hcs-c16-v1`. The mathematical arguments proving flat non-isolation, near-wall divergence, height properness and the bounded-Hecke Weyl law do not depend on the finite enumeration cutoffs.

References

[1] Shigeki Akiyama and Yoshio Tanigawa. The selberg trace formula for modular correspondences. *Nagoya Mathematical Journal*, 117:93–123, 1990.

[2] Nguyen-Thi Dang and Jialun Li. Equidistribution and counting of periodic tori in the space of weyl chambers. *Commentarii Mathematici Helvetici*, 101(1):47–113, 2026. Preprint: arXiv:2305.17070; related predecessor: arXiv:2202.08323.

[3] Anton Deitmar. The prime geodesic theorem for higher rank spaces. *Geometric and Functional Analysis*, 14:1238–1266, 2004.

[4] Anton Deitmar. Lefschetz formulae for p-adic groups. *Chinese Annals of Mathematics, Series B*, 28(4):463–474, 2007.

[5] Anton Deitmar and Ming-Hsuan Kang. Geometric zeta functions for higher rank p-adic groups. *Illinois Journal of Mathematics*, 58(3):719–738, 2014.

[6] Manfred Einsiedler and Dmitry Kleinbock. Measure rigidity and p-adic littlewood-type problems. *Compositio Mathematica*, 143(3):689–702, 2007.

[7] Ilya Khayutin. Arithmetic of double torus quotients and the distribution of periodic torus orbits. *Duke Mathematical Journal*, 168(12):2365–2432, 2019.

[8] Dmitry Kleinbock and George Tomanov. Flows on S-arithmetic homogeneous spaces and applications to metric diophantine approximation. *Commentarii Mathematici Helvetici*, 82:519–581, 2007.

[9] Pablo Ramacher and Satoshi Wakatsuki. Asymptotics for hecke eigenvalues of automorphic forms on compact arithmetic quotients. *Advances in Mathematics*, 404:108372, 2022.

[10] Yanli Song. Hecke operator and ruelle dynamical zeta function. Talk abstract, Geometry, Dynamics, and Zeta Functions, 2026. Official abstract for ongoing work; no preprint found at the 2026-08-07 source lock.

[11] John Voight. *Quaternion Algebras*, volume 288 of *Graduate Texts in Mathematics*. Springer, 2021.

[12] Liang Wang. An area-preserving hénon-map model for the riemann zeros: A deterministic-dynamics approach with quantum and dissipative solvers. Local foundational manuscript, 2026.